

Systems Verification

Exercise Sheet 03

Exercise 1: Spin's execution model

Using the properties of the *Spin* execution model, write a *Promela* model that is equivalent to the one below, but contains no `if`, `->`, or `goto` statements.

```
mtype = { P , C };

mtype turn = P;

active proctype producer(){
    wait: if
        :: (turn == P)->
            printf("Produce\n");
            turn = C;
            goto wait
        :: else -> goto wait
    fi;
}

active proctype consumer(){
    wait: if
        :: (turn == C)->
            printf("Consume\n");
            turn = P;
            goto wait
        :: else -> goto wait
    fi;
}
```

Deadline: Tuesday 15th April 2025, 8:00 a.m.

Exercise 2: Fairness and the *progress* label

Consider the following model:

```
byte x = 2;

active proctype A(){
  do
    :: x = 3-x; progress: skip;
  od
}

active proctype B(){
  do
    :: x = 3-x; progress: skip;
  od
}
```

1. When looking for non-progress cycles:
 - (a) Does Spin detect an error when using the *weak fairness* constraint? Why?
 - (b) Does Spin detect an error when is not using the weak fairness constraint? Why?
2. Since both processes are identical, we can check fairness of one process by removing one of the *progress* labels. How does Spin behave when having a progress label in only process **A** and why?
3. We now want to prove fairness for both processes. Adapt the existing processes and implement an additional process **Monitor** that handles progress labels for both original processes. Spin should detect a non-progress cycle if the weak fairness constraint is disabled.

Exercise 3: Using Message Channels

Write a *Promela* model which contains the following:

1. A synchronous channel.
2. A looping sender process that can write, at each loop, a 0 or a 1 (non-deterministically) in the channel.
3. A looping receiver process, that reads the channel and counts the number of 0's and 1's and stores the value in `counter0` and `counter1`, respectively.

Use *iSpin* to run a simulation of your model to see if it meets your expectation.

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